

PIUM — Methodological Clarification and Current Empirical Scope

Outer-Slope Baseline and the Status of the Morphological Coherence Parameter κ

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ABSTRACT

This document clarifies the current methodological status and empirical scope of the ongoing PIUM-related SPARC analysis program. The focus is on the re-analysis of SPARC galaxy rotation curves and on the introduction of a morphology–dynamics interface via the proposed coherence parameter κ .

This clarification establishes a clear separation between a reproducible kinematic baseline based on outer rotation–curve slope measurements and a currently non-robust morphological observable, κ , whose extraction remains limited by data quality and methodological constraints.

The purpose of this document is not to introduce new empirical claims, but to formalize the current status of the analysis pipeline, explicitly describe its iterative development, and define the scope of valid conclusions.

1. CONTEXT AND PURPOSE

Previous PIUM-related empirical analyses explored the possibility that large-scale galactic morphology and outer-disk kinematics are connected through a coherence-like structural parameter, denoted κ .

During the course of multiple recalculations and pipeline refinements, it became necessary to clarify which components of the analysis are currently stable and reproducible, which elements remain at the hypothesis or development stage, and how the empirical program should be interpreted in its present form.

This document addresses these points explicitly.

2. ITERATIVE DEVELOPMENT OF THE ANALYSIS

The SPARC re-analysis did not proceed as a single fixed pipeline, but evolved through several stages of refinement.

Early exploratory versions employed direct linear slope estimates over outer radial subsets, simplified definitions of the outer region, and preliminary attempts to correlate kinematic behaviour with morphological indicators.

Subsequent iterations introduced stricter sample selection criteria, normalization of radial domains, and the adoption of a log–log outer-slope estimator defined as

$$s = d \log_{10}(V_{\text{obs}}) / d \log_{10}(R),$$

measured via ordinary least-squares regression in the outer disk region.

This evolution reflects methodological refinement rather than inconsistency. The objective throughout remained the identification of a reproducible and phenomenologically interpretable descriptor of outer rotation–curve behaviour.

Earlier exploratory records may therefore contain alternative outer-slope definitions; these should be interpreted as development-stage metrics rather than the current fiducial baseline.

3. KINEMATIC BASELINE: OUTER-SLOPE MEASUREMENT

The primary reproducible outcome of the current analysis is the establishment of an outer-slope kinematic baseline.

Key properties are as follows.

The slope is defined in log–log space to ensure scale invariance.
Outer-slope estimates are obtained by ordinary least-squares regression in log–log space.
The outer region is defined as $R > 2.5 R_{\text{disk}}$.
A minimum of $N \geq 8$ data points is required in the outer region.
Only galaxies with quality flag $Q \leq 2$ are included.
The measurement procedure is reproducible across the selected SPARC subsample.

This outer-slope parameter should be regarded as the fiducial empirical quantity emerging from the present work.

No morphology-based interpretation is required for its validity.

4. STATUS OF THE MORPHOLOGICAL COHERENCE PARAMETER κ

The parameter κ was introduced as a candidate observable intended to quantify the dominance of low-order structural modes, for example $m = 2$, in galactic morphology.

Conceptually, κ is intended to represent a normalized measure of low-order azimuthal structural dominance, schematically based on radially integrated Fourier mode amplitudes.

However, in the current data environment, κ cannot yet be considered a robust empirical variable.

The main limitations are the following:

heterogeneous imaging sources in terms of resolution, wavelength, and depth,
uncertain or inconsistent deprojection parameters, including inclination and position angle,
structural ambiguity in spiral and other non-axisymmetric features,
incomplete radial coverage for reliable mode decomposition.

As a result, κ should presently be regarded as a candidate observable and candidate interface variable, not as a validated measurement.

No definitive morphology–dynamics correlation based on κ is claimed at the present stage of the analysis.

5. SEPARATION OF KINEMATIC AND MORPHOLOGICAL LIMITATIONS

It is essential to distinguish between two fundamentally different sources of uncertainty.

On the kinematic side, the main issues concern the definition of the outer region, the sampling of rotation curves, and the regression methodology.

On the morphological side, the main issues concern image quality, structural decomposition, and measurement reproducibility.

The current limitations of κ do not affect the validity of the outer-slope baseline.

6. CURRENT SCOPE OF VALID CONCLUSIONS

The present work supports the following statements:

A reproducible outer-slope descriptor can be defined for a well-selected subset of SPARC galaxies.
This descriptor provides a reproducible empirical characterization of outer-disk kinematics.
A morphology–dynamics connection remains a motivated hypothesis, but has not yet been empirically established.

The study should therefore be interpreted as the establishment of a kinematic baseline, not as a completed verification of morphology–dynamics coupling.

7. INTERPRETATION WITHIN THE PIUM FRAMEWORK

Within the PIUM framework, κ was intended to serve as a bridge between phase-coherent structural organization, expressed in morphology, and large-scale dynamical behaviour, expressed in rotation curves.

The current results neither confirm nor rule out this conceptual connection. Rather, they indicate that the observable definition of κ requires further standardization and that higher-quality, homogeneous datasets are necessary.

Thus, the PIUM-motivated empirical program remains open.

8. FUTURE DIRECTIONS

To advance the morphology–dynamics test, the following are required:

1. a standardized imaging dataset with uniform bandpass, resolution, and calibration,
2. a consistent deprojection pipeline across all galaxies,
3. a robust radial window definition for Fourier analysis,
4. explicit normalization of mode amplitudes,
5. a reproducible κ extraction protocol.

Only after these conditions are met can κ be tested as a statistical predictor of outer-slope behaviour.

Until such a pipeline is established, κ should not be used as a quantitative explanatory variable in PIUM-related empirical claims.

9. CONCLUSION

The present clarification formalizes the current status of the PIUM empirical program.

The outer-slope analysis provides a reproducible kinematic baseline.
The morphological coherence parameter κ remains a hypothesis-level observable.
The morphology–dynamics coupling remains an open empirical question.

This separation strengthens the methodological integrity of the work and defines a clear path for future investigation.

KEYWORDS

SPARC galaxies, rotation curves, outer slope, morphology–dynamics relation, Fourier decomposition, galactic structure, PIUM, empirical astrophysics

RELATED WORK

This document provides a methodological clarification of the outer rotation–curve slope analysis presented in:

<https://doi.org/10.5281/zenodo.18910574>

It does not replace the original empirical analysis, but refines its methodological definition, selection criteria, and interpretation framework.

The present work establishes a more explicit and reproducible formulation of the outer-region definition and slope measurement procedure used in SPARC-based studies.